Latihan Logistic Regression

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	x1	x2	class y
1	0.5	0	-1
2	0	-1	1
3	1	-0.5	1

Seorang mahasiswa diminta melakukan proses pembelajaran dengan Logistic Regression untuk mendapatkan bobot w0, w1, w2, w3, sebanyak 2 iterasi, jika digunakan: bobot awal w0 s.d w2 sebesar 1, nilai bias 1, learning rate 0.5. Urutan instance pembelajaran hasil pemilihan random adalah iterasi 1: 3, 1, 2; iterasi 2: 1, 3, 2.

a. Lakukan proses pelatihan, tuliskan perhitungan lengkap dengan formula dan cara perhitungan. Tuliskan bobot yang dihasilkan dari setiap iterasi.
b. Tentukan hasil prediksi kelas untuk setiap instance dengan menggunakan bobot model yang diperoleh pada no a. Hitung akurasi hasil pelatihan.

No	**Jawaban**

a

a. Proses pelatihan

- digunakan $X_0 = 1$ untuk masukan bias (sehingga bobot yang dilatih adalah w_0)
- untuk kasus rumus sigmoid: kelas $y \in \{-1, +1\}$ didekatkan menjadi target $t = \frac{(y+1)}{2}$
  - → kelas −1 jadi $t = 0$
  - → kelas +1 jadi $t = 1$
- Rumus:

$$z = w_0 + w_1 x_1 + w_2 x_2$$

$$p = \sigma(z) = \frac{1}{1+e^{-z}}$$

$$\Delta = \eta (t-p) = 0{,}5(t-p)$$

$$(w_0, w_1, w_2)_{baru} = (w_0, w_1, w_2)_{lama} + \Delta(1, x_1, x_2)$$

ITERASI dengan urutan $3 \to 1 \to 2$ (bobot awal $w = (1,1,1)$)

1) Instance 3: $(x_1, x_2) = (1, -0{,}5)$ $t = 1$

$$z = 1 + 1.1 + 1.(-0{,}5) = 1{,}5$$

$$p = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-1{,}5}} = \frac{1}{1+0{,}22313}$$

$$= 0{,}817574$$

$$\Delta = 0{,}5(1 - 0{,}817574) = 0{,}091213$$

$$w = (1,1,1) + 0{,}091213(1, 1, -0{,}5) \approx (1{,}091213;\ 1{,}091213;\ 0{,}954394)$$

2) Instance 1 $(x_1, x_2) = ((0{,}5), 0)$ $t = 0$

$$z = 1{,}091213 + 1{,}091213 \cdot (0{,}5) = 1{,}636819$$

$$p = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-1{,}636819}} = \frac{1}{1+0{,}194598}$$

$$= 0{,}837102$$

$$\Delta = 0{,}5(0 - 0{,}837102) = -0{,}418551$$

$$w = (1{,}091213;\ 1{,}091213;\ 0{,}954394) - 0{,}418551(1, (0{,}5), 0)$$

$$\approx (0{,}672662;\ 0{,}881937;\ 0{,}954394)$$

3) Instance 2 $(x_1, x_2) = (0, -1)$ $t = 1$

$$z = 0{,}672662 - 0{,}954394 = -0{,}281732$$

$$p = \frac{1}{1+e^{0{,}281732}} = 0{,}430029$$

$$\Delta = 0{,}5(1 - 0{,}430029) = 0{,}284985$$

$$w = (0{,}672662;\ 0{,}881937;\ 0{,}954394) + 0{,}284985(1, 0, -1)$$

$$\approx (0{,}957647;\ 0{,}881937;\ 0{,}669408)$$

ITERASI 2 urutan 1 → 3 → 2

1) Instance 1 $(x_1, x_2) = (0,5; 0)$ $t = 0$

$z = 0,957647 + 0,881937(0,5) = 1,398614$

$p = \frac{1}{1+e^{-1,398614}} = 0,801964$

$\Delta = 0,5(1 - 0,801964) = 0,400982$

$w = (0,957647; 0,881937; 0,669408) - 0,400982(1, (0,5), 0)$

$= (0,556605; 0,681446; 0,669408)$

2) Instance 3 $(x_1, x_2) = (1, (0,5))$ $t = 1$

$z = 0,33665 + 0,681446 - 0,669408(0,5) = 0,903407$

$p = 0,711649$

$\Delta = 0,5(1 - 0,711649) = 0,144175$

$w = (0,556605; 0,681446; 0,669408) + 0,144175(1, 1, (-0,5))$

$= (0,700841; 0,823622; 0,597321)$

3) Instance 2 $(x_1, x_2) = (0, -1)$ $t = 1$

$z = 0,900841 - 0,597321 = 0,103520$

$p = 0,525857$

$\Delta = 0,5(1 - 0,525857) = 0,237072$

$w = (0,700841; 0,823622; 0,597321) + 0,237072(1, 0, -1)$

$= (0,937912; 0,823622; 0,360249)$

b

$z = w_0 + w_1 x_1 + w_2 x_2$

$z = 0,9379 + 0,8256x_1 + 0,3603x_2$

Instance $(x_1; x_2)$	**z**	$p = \frac{1}{1+e^{-z}}$	**Prediksi**	**Class y Aktual**	**Benar /Salah**
$1(0,5; 0)$	$0,9379 + 0,8256(0,5) + 0,3603(0)$ $0,9379 + 0,4128 = 1,3507$	$\frac{1}{1+e^{-1,3507}} = 0,7943$	1	-1	S
$2(0; -1)$	$0,9379 + 0,8256(0) + 0,3603(-1)$ $0,9379 - 0,3603 = 0,5777$	$\frac{1}{1+e^{-0,5777}} = 0,6405$	1	1	B
$3(1; -0,5)$	$0,9379 + 0,8256(1) + 0,3603(-0,5)$ $0,9379 + 0,8256 - 0,1801 = 1,5834$	$\frac{1}{1+e^{-1,5834}} = 0,8297$	1	1	B

	Akurasi hasil pelatihan $= \frac{2}{3} \approx 0,6667 = 66,67\%$